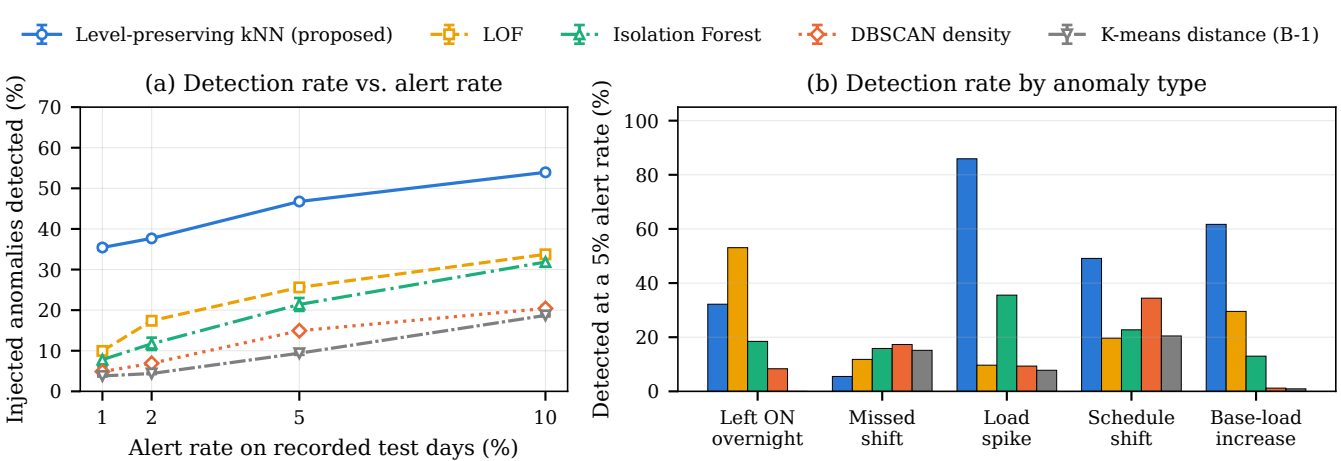
Level-preserving kNN (proposed)
LOF
Isolation Forest
DBSCAN density
K-means distance (B-1)
(a) Detection rate vs. alert rate
Injected anomalies detected (%)
Alert rate on recorded test days (%)
0
10
20
30
40
50
60
70
1
2
5
10
(b) Detection rate by anomaly type
Detected at a 5% alert rate (%)
0
20
40
60
80
100
Left ON overnight
Missed shift
Load spike
Schedule shift
Base-load increase